

- Scholes, M.S., 1972, The market for securities: Substitution versus price pressure and the effects of information on share price, *Journal of Business* 45, 179-211.
- Scholes, M.S. and J. Williams, 1977, Estimating betas from non-synchronous data, *Journal of Financial Economics* 5, 309-328.
- Schultz, P., 1983, Transactions costs and the small firm effect: A comment, *Journal of Financial Economics* 12, 81-88.
- Stoll, H.R., 1978, The supply of dealer services in securities markets, *Journal of Finance* 33, 1133-1151.
- Stoll, H.R. and R.E. Whaley, 1983, Transactions costs and the small firm effect, *Journal of Financial Economics* 12, 57-79.
- Tinic, S.M., 1972, The economics of liquidity services, *Quarterly Journal of Economics* 86, 79-93.
- Tinic, S.M. and R.R. West, 1972, Competition and the pricing of dealer service in the over-the-counter stock market, *Journal of Financial and Quantitative Analysts* 7, 1707-1728.
- Treynor, J., 1981, What does it take to win the trading game?, *Financial Analysts Journal*, 55-60.

NOTES ON ROSS' (1976) ARBITRAGE-PRICING THEORY

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Motivation -- Why do we need multifactor models?

The SLB model says that all relevant sources of uncertainty in returns that affect equilibrium expected returns can be summarized in terms of the uncertainty of next period's wealth. Investors, however, may also be concerned with other features of next period's state of the world, and they may wish to use their portfolio decisions to hedge against uncertainty about these features of the state of the world. The main examples include uncertainty about the relative prices of consumption goods, uncertainty about the investment opportunity set, for example, the level of expected returns, and uncertainty about future employment opportunities. In the multifactor models discussed below, different sources of uncertainty in returns need not have the same market price or expected risk premium. This is the fundamental distinction between the multifactor model and the one-factor SLB model.

The APT

The first model we consider is the APT (arbitrage-pricing theory) of Ross (1976). It is based on assumptions about the return-generating process. In particular, the APT assumes that the covariation of the returns on securities is captured by a small number J of common (non-diversifiable) factors in returns. The common factors are the result of the response of stock prices

to common changes in earnings prospects or discount rates for all or for subsets of firms.

Suppose the return process for any security i is a one-factor model or a multifactor model of the following form,

$$(1) \quad \tilde{R}_{it} = E(\tilde{R}_{it}) + b_i \tilde{K}_t + \tilde{\epsilon}_{it}, \quad \text{cov}(\tilde{\epsilon}_i, \tilde{\epsilon}_j) = 0, \quad j \neq i, \quad E(\tilde{K}_t) = 0,$$

$$(2) \quad \tilde{R}_{it} = E(\tilde{R}_{it}) + \sum_{j=1}^J b_{ij} \tilde{K}_{jt} + \tilde{\epsilon}_{it}, \quad \text{cov}(\tilde{\epsilon}_i, \tilde{\epsilon}_d) = 0, \quad i \neq d, \quad E(\tilde{K}_{jt}) = 0.$$

The common factors, the $\tilde{K}$ s in (1) and (2), can always be transformed so that, as assumed, they have means equal to 0. The b 's in factor models, are called factor sensitivities or factor loadings. They can be interpreted as simple regression slopes for (1) and multiple regression slopes for (2). Equations (1) and (2) then say that the unexpected return on security i has one or more common factors that are shared with other securities, as well as an idiosyncratic or firm-specific residual return.

If there is an infinite number of securities and each security is an infinitesimal fraction of total wealth, it is possible to form perfectly diversified portfolios, that is, portfolios that have no residual error term, $\tilde{\epsilon}_{pt}=0$. For example, in the one-factor model (1), and for an equally weighted portfolio of n securities,

$$(3a) \quad \tilde{R}_{pt} = \sum_{i=1}^n x_{ip} \tilde{R}_{it} = \sum_{i=1}^n x_{ip} [E(\tilde{R}_{it}) + b_i \tilde{K}_t + \tilde{\epsilon}_{it}],$$

$$(3b) \quad = E(\tilde{R}_{pt}) + b_p \tilde{K}_t + \sum_{i=1}^n x_{ip} \tilde{\epsilon}_{it},$$

$$(3c) \quad = E(\tilde{R}_{pt}) + b_p \tilde{K}_t + (1/n) \sum_{i=1}^n \tilde{\epsilon}_{it}.$$

The variance of the return on this equally weighted portfolio is then,

$$(4a) \quad \sigma^2(\tilde{R}_p) = b_p^2 \sigma^2(\tilde{K}) + (1/n)^2 \sum_{i=1}^n \sigma^2(\tilde{\epsilon}_i)$$

$$(4b) \quad \rightarrow b_p^2 \sigma^2(\tilde{K}) \quad \text{as } n \rightarrow \infty.$$

In other words, in a perfectly diversified portfolio, firm-specific variance is diversified away; there is no error term,

$$(5) \quad \tilde{R}_{pt} = E(\tilde{R}_{pt}) + b_p \tilde{K}_t;$$

Similarly, in the multifactor model, a perfectly diversified portfolio implies,

$$(6) \quad \tilde{R}_{pt} = E(\tilde{R}_{pt}) + \sum_{j=1}^J b_{pj} \tilde{K}_{jt}.$$

A perfectly diversified portfolio does not necessarily imply equal weights for individual securities. But the weight invested in each security must approach 0 as $n \rightarrow \infty$. If the weight on a security does not approach 0, the contribution of that security's residual variance to the variance of the return on the portfolio does not approach 0.

The market-equilibrium argument in the APT is just a no-arbitrage condition. In a market equilibrium there cannot be riskless arbitrage opportunities. The APT says that this no-arbitrage condition implies that there is a linear relation between the expected returns on perfectly diversified portfolios and their factor loadings, the b 's. Specifically, the APT says that for the one-factor and multifactor models, respectively,

$$(7) \quad E(\tilde{R}_{pt}) = R_{ft} + b_p [E(\tilde{R}_{kt}) - R_{ft}],$$

$$(8) \quad E(\tilde{R}_{pt}) = R_{ft} + \sum_{j=1}^J b_{pj} [E(\tilde{R}_{jt}) - R_{ft}].$$

In the one-factor model, $E(R_{kt})$ is the expected return on a perfectly diversified portfolio that has $b_k=1$. In the multifactor model, $E(\tilde{R}_{jt})$ is the expected return on a perfectly diversified portfolio that has $b_j=1$ and $b_i=0$, $i \neq j$.

The Argument (One-Factor model)

Consider the one-factor model first. Given unrestricted short-selling, if there are two perfectly diversified portfolios with different b 's, it is possible to use them to create perfectly diversified portfolios with any value of b . This means it is possible to create a perfectly diversified portfolio with $b=1$, call it portfolio k . It is also possible to create a perfectly diversified portfolio with $b=0$. From (5) this portfolio is riskless; that is, the variance of its return is 0. Thus in the APT, with unrestricted short-selling, a riskless portfolio, call it portfolio f , exists.

Alternatively, one can simply assume that there is a risk-free security f , and there is unrestricted risk-free borrowing and lending. Then, given unrestricted short-selling, if there is one perfectly diversified portfolio with a non-zero b , it is possible to combine this portfolio with risk-free borrowing or lending to create perfectly diversified portfolios with any value of b . This means that it is possible to create a perfectly diversified portfolio with $b=1$; call it portfolio k .

It is clear from (5) that two perfectly diversified portfolios with the same b must have the same expected return. If they did not, a sure (zero variance) arbitrage profit could be

obtained by shorting \$1 of the lower expected return portfolio and putting the \$1 in the higher expected return portfolio. Since the two portfolios are perfectly diversified and they have the same b , (5) says that the arbitrage portfolio has zero variance and a sure profit, which is equal to the difference between the expected returns on the two portfolios.

Now the punch line. Consider an arbitrary perfectly diversified portfolio p . Thus,

$$(9) \quad \tilde{R}_{pt} = E(\tilde{R}_{pt}) + b_p \tilde{K}_t.$$

A combination of the risk-free portfolio f and the perfectly diversified portfolio k that has $b_k=1$ can always be used to form a perfectly diversified portfolio that has $b=b_p$. Specifically, form the portfolio q that invests b_p in k and $1-b_p$ in f . The return on this portfolio is,

$$\begin{aligned} \tilde{R}_{qt} &= b_p \tilde{R}_{kt} + (1-b_p) R_{ft} \\ (10) \quad &= b_p [E(\tilde{R}_{kt}) + \tilde{K}_t] + (1-b_p) R_{ft} \\ &= R_{ft} + b_p [E(\tilde{R}_{kt}) - R_{ft}] + b_p \tilde{K}_t. \end{aligned}$$

Since the portfolios p and q are perfectly diversified and they have the same b , they must have the same expected return,

$$\begin{aligned} (11) \quad E(\tilde{R}_{pt}) &= E(\tilde{R}_{qt}) \\ &= R_{ft} + b_p [E(\tilde{R}_{kt}) - R_{ft}]. \end{aligned}$$

Thus we have shown that the expected return on any perfectly diversified portfolio must satisfy (7). In short, for perfectly diversified portfolios, the no-arbitrage condition implies that there must be an exact linear relation between expected returns and their loadings on the one common factor in returns.

The Argument (Multifactor model)

The argument generalizes to the multifactor model. Given that there are J factors, suppose there are at least $J+1$ perfectly diversified portfolios and every subset J of them has linearly independent factor loading vectors. Then, using these $J+1$ portfolios, it is possible to create J perfectly diversified portfolios that each has a $b_j=1$ and all other $b_j=0$. [We need the $J+1$ st portfolio to ensure that when we combine the portfolios, the weights on the $S+1$ portfolios sum to 1.0.] The returns on these J portfolios are,

$$(12) \quad \tilde{R}_{jt} = E(\tilde{R}_{jt}) + \tilde{K}_{jt}, \quad j=1, \dots, J,$$

where $\tilde{R}_{jt}$ is the return on the perfectly diversified portfolio that has $b_{pj}=1$ and all other b 's equal to 0.

Note that the uncertainty about $\tilde{R}_{jt}$ is just the common factor $\tilde{K}_{jt}$. Thus the factor loadings b_{ij} , $j=1..J$, in (2), are also the slopes in the time-series multiple regression of the return on security i on the returns $\tilde{R}_{jt}$, $j=i..J$, on the perfectly diversified portfolios that mimic the j common factors in returns. In other words, the $\tilde{R}_{jt}$ will serve as well as the $\tilde{K}_{jt}$ in the factor model (2) and in regressions to determine factor loadings.

If there are at least $J+1$ perfectly diversified portfolios and every subset J of them has linearly independent factor loading vectors, it is also possible to create a riskless portfolio, that is, a perfectly diversified portfolio (call it f) that has all $b_j=0$. Alternatively, one can simply assume that

there is unrestricted risk-free borrowing and lending at the rate R_{ft} .

Now consider any perfectly diversified portfolio p that possibly has non-zero b 's on all factors. The return on this portfolio is,

$$(13) \quad \tilde{R}_{pt} = E(\tilde{R}_{pt}) + \sum_{j=1}^J b_{pj} \tilde{K}_{jt}.$$

We can form another perfectly diversified portfolio that replicates the b_{pj} 's of portfolio p by investing b_{pj} in the J factor portfolios that have $b_j=1$ and zero b 's on other factors, and then investing $1 - \sum_j b_{pj}$ in the risk-free portfolio. The return on the resulting portfolio, call it q , will be,

$$\begin{aligned} \tilde{R}_{qt} &= \sum_{j=1}^J b_{pj} \tilde{R}_{jt} + (1 - \sum_{j=1}^J b_{pj}) R_{ft} \\ (14) \quad &= \sum_{j=1}^J b_{pj} E(\tilde{R}_{jt}) + \sum_{j=1}^J b_{pj} \tilde{K}_{jt} + (1 - \sum_{j=1}^J b_{pj}) R_{ft} \\ &= R_{ft} + \sum_{j=1}^J b_{pj} [E(\tilde{R}_{jt}) - R_{ft}] + \sum_{j=1}^J b_{pj} \tilde{K}_{jt}. \end{aligned}$$

The no-arbitrage condition implies that since (a) the portfolios p and q are perfectly diversified and (b) they have the same loadings on the J factors, then p and q must have the same expected return,

$$\begin{aligned} E(\tilde{R}_{pt}) &= E(\tilde{R}_{qt}) \\ (15) \quad &= R_{ft} + \sum_{j=1}^J b_{pj} [E(\tilde{R}_{jt}) - R_{ft}]. \end{aligned}$$

Thus we have shown that the expected return on any perfectly diversified portfolio must satisfy (8). In short, for perfectly diversified portfolios, the no-arbitrage condition implies that

there must be an exact linear relation between expected returns and their loadings on the common factors in returns.

What about individual securities? We can't guarantee that (8) applies to every individual security because the arbitrage argument applies only to perfectly diversified portfolios. In a perfectly diversified portfolio, the weight on an individual security, or on any finite subset of securities, is arbitrarily close to 0. Thus, (8) can be violated for a finite subset of securities [specifically, the expected returns on securities in this subset can differ from (8) by large but strictly bounded quantities], and the subset can include a large number of securities. Some argue that this makes the APT empirically empty.

The Multiperiod Intertemporal Asset-Pricing Model

Fortunately, Merton (1973) derives a risk-return relation like (8) in an intertemporal model using assumptions about tastes (basically risk aversion) and portfolio opportunities (a kind of multivariate normality). In his model, (8) holds for all securities and portfolios. The one-factor version of the Merton model is the SLB model. The multifactor version allows for additional factors that capture the incentives of investors to hedge against different specific sources of uncertainty about future consumption, investment, and employment opportunities (called future state variables).

REFERENCES

- Merton, Robert C., 1973, An intertemporal capital asset pricing model, *Econometrica*, 41, 867-887.
- Ross, Stephen A., 1976, The arbitrage theory of capital asset pricing, *Journal of Economic Theory* 13, 341-60.